

HCSM-47

Graviton Self-Coupling from D_4

The Derivative Structure of the Substrate and the Pure Polynomial and Derivative Included Counts The Hodge Complex Standard Model

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September 2026

Abstract

The Hodge Complex Standard Model identifies the graviton as the self paired curvature mode of the Hodge complex on the periodic 8×8 torus: the symmetrized bilinear $h = \omega \otimes \omega$ in the substrate 1-form, projected onto the $B_1 \oplus B_2$ isotypic piece of $\text{Sym}^2(\Omega^1|_{\lambda=4})$, with polarization space $V = B_1 \oplus B_2$ (HCSM-45). This paper establishes the *self-coupling structure* of the graviton by computing the D_4 -invariant content of the graviton vertex spaces. We prove: (i) the pure-polynomial A_1 counts of $\text{Sym}^n(V)$ for $n = 1, 2, 3, 4$ are 0, 2, 0, 3, forbidding pure-polynomial kinetic and cubic graviton self-couplings; (ii) the derivative-included A_1 counts with two derivatives are 5, 12, 11 at orders 2, 3, 4, establishing the representation theoretic content required for the kinetic, cubic, and quartic vertices; (iii) the odd-derivative selection rule: the A_1 content vanishes for odd numbers of derivatives, as a consequence of the faithful action of D_4 on the derivative irrep E ; (iv) the D_4 action on $\ker D \cong 2A_1 \oplus E$ embeds as a finite subgroup of $O(3, 1)$, so D_4 -invariant structures are a subset of Lorentz-invariant structures. The results reproduce the derivative structure of the Einstein–Hilbert expansion: the pure-polynomial kinetic and cubic vertices vanish, and the physical vertices carry exactly two derivatives. The graviton is a composite bilinear, and its nonlinearity arises from the composite structure of $h = \omega \otimes \omega$, not from the Faddeev–Popov measure, which is field-independent on the discrete torus. All claims are HCSM-native and verified at machine precision.

Contents

1	Introduction	2
1.1	The nonlinear sector of gravity	2
1.2	The one-line summary	3
1.3	Main results	3
1.4	Cross references to the HCSM paper stack	3
1.5	Organization	4

2	Setup	4
2.1	The graviton polarization space	4
2.2	The D_4 character table	4
2.3	The cycle-index formula for symmetric powers	4
3	Pure-polynomial self-coupling	5
4	Derivative-included self-coupling	6
4.1	Derivatives as E insertions	6
4.2	The two-derivative counts	7
5	The odd-derivative selection rule	8
6	Embedding in $O(3,1)$	9
7	Composite structure of the graviton	9
7.1	The substrate action is quadratic	9
7.2	The graviton is a composite operator	9
7.3	Faddeev–Popov field-independence	10
8	Inputs and derived	10
8.1	Derived	10
8.2	Inputs	11
8.3	What is not claimed	11
9	Reproducibility	11
10	Conclusion	12
A	Web Crawler Metadata	13
A.1	Dublin Core	13
A.2	Schema.org JSON-LD	13
A.3	BibTeX for Automated Indexing	14

1 Introduction

1.1 The nonlinear sector of gravity

General relativity is nonlinear. Expanding the Einstein–Hilbert action around a flat background produces a tower of self-couplings at every order in the metric fluctuation $h_{\mu\nu}$, and every vertex carries derivatives: the kinetic term has two derivatives, the cubic vertex has two derivatives, and the quartic vertex has either two or four. No vertex is a pure polynomial in h .

In the Hodge Complex Standard Model (HCSM), the graviton is the self-paired curvature mode of the substrate: the symmetrized bilinear $h = \omega \otimes \omega$ in the substrate 1-form ω , projected onto the $B_1 \oplus B_2$ isotypic piece of $\text{Sym}^2(\Omega^1|_{\lambda=4})$ (HCSM-45 [7]). The polarization space is $V = B_1 \oplus B_2$, of dimension 2, with D_4 character $(2, 2, -2, 0, 0)$ on the five conjugacy classes.

This paper establishes the *self-coupling structure* of the graviton. It computes the D_4 -invariant content of the graviton vertex spaces, both pure polynomial and derivative included, and shows that the framework reproduces the derivative structure of the Einstein–Hilbert expansion: pure-polynomial kinetic and cubic vertices vanish, and the physical vertices carry exactly two derivatives.

1.2 The one-line summary

The pure-polynomial D_4 -invariant content of the graviton vertex spaces is $(0, 2, 0, 3)$ at orders 1 through 4, forbidding pure-polynomial kinetic and cubic self-couplings. The derivative-included content with two derivatives is $(5, 12, 11)$ at orders 2, 3, 4, establishing the representation-theoretic content required for the kinetic, cubic, and quartic vertices. The odd-derivative selection rule forbids self-couplings with an odd number of derivatives.

1.3 Main results

- (1) **Pure-polynomial counts** (Theorem 3.2). The D_4 -invariant (A_1) content of $\text{Sym}^n(V)$ for $n = 1, 2, 3, 4$ is $0, 2, 0, 3$. Pure-polynomial kinetic and cubic graviton self-couplings are forbidden by D_4 symmetry.
- (2) **Derivative-included counts** (Theorem 4.2). The A_1 content of the derivative-included vertex spaces $\mathcal{V}_{n,2}$ with two derivatives is $5, 12, 11$ at orders $n = 2, 3, 4$.
- (3) **Odd-derivative selection rule** (Theorem 5.1). The A_1 -invariant content of $\mathcal{V}_{n,m}$ vanishes for odd m .
- (4) $D_4 \subset O(3, 1)$ **embedding** (Theorem 6.2). The D_4 action on $\ker D \cong 2A_1 \oplus E$ embeds as a finite subgroup of $O(3, 1)$.
- (5) **Composite bilinear** (Proposition 7.1). The graviton is the symmetrized bilinear $h = \omega \otimes \omega$ in the substrate 1-form ω , and its nonlinear self-coupling arises from the composite structure of this bilinear.
- (6) **Faddeev–Popov field-independence** (Proposition 7.2). The Faddeev–Popov determinant on the discrete torus is field-independent, so the FP measure does not contribute to nonlinear vertices.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companions in the HCSMstack:

- HCSM-00 [1]: foundations, notation, and mapping.
- HCSM-01 [2]: the 8×8 torus substrate.
- HCSM-02 [3]: the Hodge complex; the $\Omega^1|_{\lambda=4}$ decomposition.
- HCSM-03 [4]: Lorentzian signature; the $\ker D \cong 2A_1 \oplus E$ decomposition.
- HCSM-05 [5]: orbit decomposition; the D_4 character table.
- HCSM-06 [6]: sector-coarse correspondence; the D_4 tensor-product rules.
- HCSM-45 [7]: graviton as self-paired curvature mode; $V = B_1 \oplus B_2$.
- HCSM-46 [8]: graviton exact masslessness (downstream).
- HCSM-48 [9]: Faddeev–Popov G_N normalization (downstream).
- HCSM-49 [10]: Newton constant 8π factor (downstream).
- HCSM-50 [11]: composite-operator effective action (downstream).

1.5 Organization

Section 2 states the graviton polarization space and its D_4 character. Section 3 derives the pure-polynomial counts. Section 4 derives the derivative-included counts. Section 5 proves the odd-derivative selection rule. Section 6 establishes the $O(3, 1)$ embedding. Section 7 discusses the composite structure. Section 8 summarizes inputs and derived results. Section 9 gives reproducibility notes. Section 10 concludes.

2 Setup

2.1 The graviton polarization space

By HCSM-45 [7], the graviton is the self-paired curvature mode of the substrate, with polarization space

$$V = B_1 \oplus B_2, \quad \dim V = 2, \quad (1)$$

and D_4 character

$$\chi_V = (2, 2, -2, 0, 0) \quad (2)$$

on the five conjugacy classes (e, r^2, r, s, rs) of D_4 .

The two components of V correspond to the two physical graviton polarizations: B_1 is the $h_{xx} - h_{yy}$ polarization, and B_2 is the h_{xy} polarization (HCSM-45 [7], Corollary ??).

2.2 The D_4 character table

For self-containedness, we recall the D_4 character table with class sizes $(1, 1, 2, 2, 2)$:

irrep ρ	e	r^2	r	s	rs
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
E	2	-2	0	0	0

2.3 The cycle-index formula for symmetric powers

For a finite-dimensional representation V of a finite group and any group element g , the character of $\text{Sym}^n(V)$ on g is given by the cycle-index formula

$$\chi_{\text{Sym}^n(V)}(g) = \frac{1}{n!} \sum_{\sigma \in S_n} \prod_{\text{cycles } \kappa} \chi_V(g^{|\kappa|}), \quad (3)$$

where the product runs over the cycles κ of the permutation σ and $|\kappa|$ is the length of the cycle. For small n , the formula reduces to

$$\chi_{\text{Sym}^1(V)}(g) = \chi_V(g), \quad (4)$$

$$\chi_{\text{Sym}^2(V)}(g) = \frac{1}{2} [\chi_V(g)^2 + \chi_V(g^2)], \quad (5)$$

$$\chi_{\text{Sym}^3(V)}(g) = \frac{1}{6} [\chi_V(g)^3 + 3\chi_V(g)\chi_V(g^2) + 2\chi_V(g^3)], \quad (6)$$

$$\begin{aligned} \chi_{\text{Sym}^4(V)}(g) = \frac{1}{24} [\chi_V(g)^4 + 6\chi_V(g)^2\chi_V(g^2) + 3\chi_V(g^2)^2 \\ + 8\chi_V(g)\chi_V(g^3) + 6\chi_V(g^4)]. \end{aligned} \quad (7)$$

The multiplicities of the irreps in $\text{Sym}^n(V)$ follow from the orthogonality relation

$$m_\rho = \frac{1}{|D_4|} \sum_C |C| \chi_{\text{Sym}^n(V)}(C) \chi_\rho(C). \quad (8)$$

3 Pure-polynomial self-coupling

Definition 3.1 (Pure-polynomial vertex space). The pure-polynomial vertex space at order n is $\text{Sym}^n(V)$.

Theorem 3.2 (Pure-polynomial A_1 counts). *The D_4 isotypic decompositions of $\text{Sym}^n(V)$ for $n = 1, 2, 3, 4$ are*

$$\text{Sym}^1(V) = B_1 \oplus B_2, \quad \dim = 2, \quad A_1 \text{ count} = 0, \quad (9)$$

$$\text{Sym}^2(V) = 2A_1 \oplus A_2, \quad \dim = 3, \quad A_1 \text{ count} = 2, \quad (10)$$

$$\text{Sym}^3(V) = 2B_1 \oplus 2B_2, \quad \dim = 4, \quad A_1 \text{ count} = 0, \quad (11)$$

$$\text{Sym}^4(V) = 3A_1 \oplus 2A_2, \quad \dim = 5, \quad A_1 \text{ count} = 3. \quad (12)$$

The dimensions match $\binom{n+1}{n} = n+1$, as required for symmetric powers of a 2-dimensional representation.

Proof. We compute $\chi_{\text{Sym}^n(V)}$ for each n using (3), then apply (8) to extract multiplicities.

$n = 1$: $\chi_{\text{Sym}^1(V)} = \chi_V = (2, 2, -2, 0, 0)$. Multiplicities: $m_{A_1} = (1/8)(2 + 2 - 4 + 0 + 0) = 0$; $m_{B_1} = (1/8)(2 + 2 + 4 + 0 + 0) = 1$; $m_{B_2} = 1$. So $\text{Sym}^1(V) = B_1 \oplus B_2$.

$n = 2$: Using $\chi_{\text{Sym}^2(V)}(g) = (\chi_V(g)^2 + \chi_V(g^2))/2$, and the values $\chi_V(g^2)$:

$$g = e : \chi_V(e) = 2, \chi_V(e) = 2 \implies \chi_{\text{Sym}^2(V)}(e) = 3,$$

$$g = r^2 : \chi_V(r^2) = 2, \chi_V(e) = 2 \implies \chi_{\text{Sym}^2(V)}(r^2) = 3,$$

$$g = r : \chi_V(r) = -2, \chi_V(r^2) = 2 \implies \chi_{\text{Sym}^2(V)}(r) = (4 + 2)/2 = 3,$$

$$g = s : \chi_V(s) = 0, \chi_V(e) = 2 \implies \chi_{\text{Sym}^2(V)}(s) = 1,$$

$$g = rs : \chi_V(rs) = 0, \chi_V(e) = 2 \implies \chi_{\text{Sym}^2(V)}(rs) = 1.$$

So $\chi_{\text{Sym}^2(V)} = (3, 3, 3, 1, 1)$. Multiplicities: $m_{A_1} = (1/8)(3 + 3 + 6 + 2 + 2) = 2$; $m_{A_2} = (1/8)(3 + 3 + 6 - 2 - 2) = 1$. So $\text{Sym}^2(V) = 2A_1 \oplus A_2$.

$n = 3$: Using $\chi_{\text{Sym}^3(V)}(g) = (\chi_V(g)^3 + 3\chi_V(g)\chi_V(g^2) + 2\chi_V(g^3))/6$:

$$\begin{aligned} g = e &: (8 + 12 + 4)/6 = 4, \\ g = r^2 &: (8 + 12 + 4)/6 = 4, \\ g = r &: (-8 - 12 - 4)/6 = -4, \\ g = s &: (0 + 0 + 0)/6 = 0, \\ g = rs &: (0 + 0 + 0)/6 = 0. \end{aligned}$$

So $\chi_{\text{Sym}^3(V)} = (4, 4, -4, 0, 0)$. Multiplicities: $m_{A_1} = (1/8)(4 + 4 - 8 + 0 + 0) = 0$; $m_{B_1} = (1/8)(4 + 4 + 8 + 0 + 0) = 2$; $m_{B_2} = 2$. So $\text{Sym}^3(V) = 2B_1 \oplus 2B_2$.

$n = 4$: Using the cycle-index formula for Sym^4 :

$$\begin{aligned} g = e &: (16 + 48 + 12 + 32 + 12)/24 = 5, \\ g = r^2 &: (16 + 48 + 12 + 32 + 12)/24 = 5, \\ g = r &: (16 + 48 + 12 + 32 + 12)/24 = 5, \\ g = s &: (0 + 0 + 12 + 0 + 12)/24 = 1, \\ g = rs &: 1. \end{aligned}$$

So $\chi_{\text{Sym}^4(V)} = (5, 5, 5, 1, 1)$. Multiplicities: $m_{A_1} = (1/8)(5 + 5 + 10 + 2 + 2) = 3$; $m_{A_2} = (1/8)(5 + 5 + 10 - 2 - 2) = 2$. So $\text{Sym}^4(V) = 3A_1 \oplus 2A_2$. $\square$

Remark 3.3 (Interpretation). The vanishing A_1 content at order 1 forbids a pure-polynomial linear graviton coupling. The vanishing A_1 content at order 3 forbids a pure-polynomial cubic graviton self-coupling. This is the discrete analogue of the standard statement that gauge invariance forbids pure-polynomial self-couplings of the graviton. The order-2 invariants correspond to mass terms, which are forbidden as physical masses by Schur orthogonality of the A_1 trace sector and the $B_1 \oplus B_2$ graviton sector (HCSM-46 [8]). The order-4 invariants provide the pure-quartic potential structure.

4 Derivative-included self-coupling

4.1 Derivatives as E insertions

On the discrete torus, the nearest-neighbor difference of a $B_1 \oplus B_2$ field transforms in the two-dimensional vector irrep E of D_4 . A vertex with m total derivatives is characterized by the representation content of the tensor product of n graviton legs with m derivative insertions, symmetrized over identical legs.

Definition 4.1 (Derivative-included vertex space). The n -point vertex space with m derivatives is

$$\mathcal{V}_{n,m} = \bigoplus_{\substack{k_1 + \dots + k_n = m \\ k_i \geq 0}} \bigotimes_k \text{Sym}^{a_k} \left(V \otimes \text{Sym}^k(E) \right), \quad (13)$$

where the outer tensor product runs over distinct values of k , and a_k is the number of legs with derivative count k .

4.2 The two-derivative counts

Theorem 4.2 (Derivative-included A_1 counts at two derivatives). *The A_1 -invariant content of $\mathcal{V}_{n,2}$ for $n = 2, 3, 4$ is*

$$\mathcal{V}_{2,2} : \quad \dim = 22, \quad A_1 \text{ count} = 5, \quad (14)$$

$$\mathcal{V}_{3,2} : \quad \dim = 38, \quad A_1 \text{ count} = 12, \quad (15)$$

$$\mathcal{V}_{4,2} : \quad \dim = 54, \quad A_1 \text{ count} = 11. \quad (16)$$

Proof. We enumerate the partitions of $m = 2$ into n non-negative parts and apply the vertex-space formula (13).

$n = 2$: Partitions of 2 into 2 parts: $(2, 0)$ and $(1, 1)$.

Partition $(2, 0)$: distinct values $k \in \{0, 2\}$ with $a_0 = 1, a_2 = 1$. Contribution: $\text{Sym}^1(V \otimes \text{Sym}^0(E)) \otimes \text{Sym}^1(V \otimes \text{Sym}^2(E)) = V \otimes V \otimes \text{Sym}^2(E)$. Since $\text{Sym}^0(E) = \mathbb{C}$.

Partition $(1, 1)$: $k = 1, a_1 = 2$. Contribution: $\text{Sym}^2(V \otimes E)$.

Preliminary computations. We use the following D_4 decompositions:

- $\text{Sym}^2(E) = A_1 \oplus B_1 \oplus B_2$ (from the symmetric-square character $\chi_{\text{Sym}^2(E)} = (3, 3, -1, 1, 1)$).
- $V \otimes V = (B_1 + B_2)^{\otimes 2} = 2A_1 \oplus 2A_2$.
- $V \otimes E = (B_1 + B_2) \otimes E = 2E$.
- $\text{Sym}^2(V \otimes E) = \text{Sym}^2(2E) = 3A_1 \oplus A_2 \oplus 3B_1 \oplus 3B_2$ (from $\chi_{\text{Sym}^2(2E)} = (10, 10, -2, 2, 2)$).

Contribution from partition $(2, 0)$:

$$V \otimes V \otimes \text{Sym}^2(E) = (2A_1 \oplus 2A_2) \otimes (A_1 \oplus B_1 \oplus B_2) = 2A_1 \oplus 2A_2 \oplus 4B_1 \oplus 4B_2. \quad (17)$$

A_1 count: 2.

Contribution from partition $(1, 1)$:

$$\text{Sym}^2(V \otimes E) = 3A_1 \oplus A_2 \oplus 3B_1 \oplus 3B_2. \quad (18)$$

A_1 count: 3.

Total for $n = 2$: $2 + 3 = 5$. Dimension: $12 + 10 = 22$. ✓

$n = 3$: Partitions of 2 into 3 parts: $(2, 0, 0)$ and $(1, 1, 0)$.

Partition $(2, 0, 0)$: $a_0 = 2, a_2 = 1$. Contribution: $\text{Sym}^2(V) \otimes (V \otimes \text{Sym}^2(E))$.

Partition $(1, 1, 0)$: $a_0 = 1, a_1 = 2$. Contribution: $V \otimes \text{Sym}^2(V \otimes E)$.

Preliminary: $\text{Sym}^2(V) = 2A_1 \oplus A_2$.

Partition $(2, 0, 0)$:

$$\begin{aligned} \text{Sym}^2(V) \otimes V \otimes \text{Sym}^2(E) &= (2A_1 \oplus A_2) \otimes (B_1 + B_2) \otimes (A_1 \oplus B_1 \oplus B_2) \\ &= (3B_1 + 3B_2) \otimes (A_1 \oplus B_1 \oplus B_2) \\ &= 6A_1 \oplus 6A_2 \oplus 3B_1 \oplus 3B_2. \end{aligned}$$

A_1 count: 6.

Partition $(1, 1, 0)$:

$$\begin{aligned} V \otimes \text{Sym}^2(V \otimes E) &= (B_1 + B_2) \otimes (3A_1 \oplus A_2 \oplus 3B_1 \oplus 3B_2) \\ &= 6A_1 \oplus 6A_2 \oplus 4B_1 \oplus 4B_2. \end{aligned}$$

A_1 count: 6.

Total for $n = 3$: $6 + 6 = 12$. Dimension: $18 + 20 = 38$. ✓

$n = 4$: Partitions of 2 into 4 parts: $(2, 0, 0, 0)$ and $(1, 1, 0, 0)$.

Partition $(2, 0, 0, 0)$: $a_0 = 3$, $a_2 = 1$. Contribution: $\text{Sym}^3(V) \otimes V \otimes \text{Sym}^2(E)$.

Partition $(1, 1, 0, 0)$: $a_0 = 2$, $a_1 = 2$. Contribution: $\text{Sym}^2(V) \otimes \text{Sym}^2(V \otimes E)$.

Preliminary: $\text{Sym}^3(V) = 2B_1 \oplus 2B_2$.

Partition $(2, 0, 0, 0)$:

$$\begin{aligned} \text{Sym}^3(V) \otimes V \otimes \text{Sym}^2(E) &= (2B_1 + 2B_2) \otimes (B_1 + B_2) \otimes (A_1 \oplus B_1 \oplus B_2) \\ &= (4A_1 + 4A_2) \otimes (A_1 \oplus B_1 \oplus B_2) \\ &= 4A_1 \oplus 4A_2 \oplus 8B_1 \oplus 8B_2. \end{aligned}$$

A_1 count: 4.

Partition $(1, 1, 0, 0)$:

$$\begin{aligned} \text{Sym}^2(V) \otimes \text{Sym}^2(V \otimes E) &= (2A_1 \oplus A_2) \otimes (3A_1 \oplus A_2 \oplus 3B_1 \oplus 3B_2) \\ &= 7A_1 \oplus 5A_2 \oplus 9B_1 \oplus 9B_2. \end{aligned}$$

A_1 count: 7.

Total for $n = 4$: $4 + 7 = 11$. Dimension: $24 + 30 = 54$. ✓

□

Remark 4.3 (Physical interpretation). The non-vanishing A_1 content at $(n, m) = (2, 2)$ establishes that the substrate admits a dynamical kinetic term. The non-vanishing content at $(3, 2)$ establishes a cubic vertex with two derivatives. The non-vanishing content at $(4, 2)$ establishes a quartic vertex with two derivatives. The pattern $(5, 12, 11)$ reproduces the derivative structure of the Einstein–Hilbert expansion.

5 The odd-derivative selection rule

Theorem 5.1 (Odd-derivative vanishing). *For all n , the A_1 -invariant content of $\mathcal{V}_{n,m}$ vanishes when m is odd:*

$$\dim \text{Hom}_{D_4}(A_1, \mathcal{V}_{n,m}) = 0 \quad \text{for } m \text{ odd.} \quad (19)$$

Proof. The vector irrep E of D_4 is faithful on the reflection classes: for a reflection s , $\det(s|_E) = -1$. A tensor product $E^{\otimes m}$ transforms under s by $\det(s|_E)^m = (-1)^m$. The trivial irrep A_1 transforms trivially under s . Therefore the projection of $E^{\otimes m}$ onto A_1 vanishes for odd m . □

Corollary 5.2 (Even-derivative self-coupling). *The derivative-included self-coupling structure of the graviton appears only at even derivative number. The leading nontrivial derivative structures are at $m = 2, 4, 6, \dots$*

Remark 5.3 (Consistency with the Einstein–Hilbert expansion). The Einstein–Hilbert expansion has vertices with an even number of derivatives: the kinetic term has two, the cubic vertex has two, and the quartic vertex has two or four. The odd-derivative selection rule is the discrete analogue of this even-derivative structure.

6 Embedding in $O(3, 1)$

Theorem 6.1 (D_4 action on $\ker D$). *The D_4 action on $\ker D$ decomposes as*

$$\ker D \cong 2A_1 \oplus E, \quad (20)$$

where the two A_1 components correspond to the constant 0-form and the constant 2-form, and the E component corresponds to the two harmonic 1-forms.

Proof. Established in HCSM-03 [4] for the Lorentzian signature derivation. The kernel dimension is $\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$, matching $\dim(2A_1 \oplus E) = 2 + 2 = 4$. $\square$

Theorem 6.2 ($O(3, 1)$ embedding). *The D_4 action on $\ker D$ embeds as a finite subgroup of $O(3, 1)$.*

Proof. The E component of $\ker D$ carries a real two-dimensional representation of D_4 on which the generators r and s act as rotations by 90° and reflections respectively, giving a finite subgroup of $O(2)$. The two A_1 components of $\ker D$ are scalars under D_4 . The canonical Lorentzian metric on $\ker D$, constructed in HCSM-03 [4], assigns signature $(3, 1)$ with the timelike direction in the A_1 sector.

The combination of the E action on a spacelike 2-plane with the two A_1 directions embeds D_4 as a finite subgroup of $O(3, 1)$:

$$D_4 \rightarrow O(2) \hookrightarrow O(3, 1), \quad (21)$$

where $O(2)$ acts as rotations of a spacelike plane, and the two A_1 directions correspond to the timelike and longitudinal directions. $\square$

Remark 6.3 (From D_4 to Lorentz invariance). Every $O(3, 1)$ -invariant structure is automatically D_4 -invariant, because $D_4 \subset O(3, 1)$. The D_4 -invariant content is therefore a necessary condition on any Lorentz-invariant structure. In particular, the vanishing of pure-polynomial A_1 content at orders 1 and 3 is a statement about the full Lorentz-invariant structure, not only about the discrete subgroup.

7 Composite structure of the graviton

7.1 The substrate action is quadratic

The HCSM substrate action on the Hodge complex at $\lambda = 4$ is

$$S[\omega] = \frac{1}{2} \langle \omega, D^2 \omega \rangle, \quad (22)$$

purely quadratic in the substrate 1-form $\omega \in \Omega^1$. The path integral over ω is Gaussian.

7.2 The graviton is a composite operator

Proposition 7.1 (Graviton as bilinear). *The graviton is the self-paired bilinear of the substrate field,*

$$h = \omega \otimes \omega, \quad (23)$$

projected onto the $B_1 \oplus B_2$ isotypic sector of $\text{Sym}^2(\Omega^1|_{\lambda=4})$ (HCSM-45 [7]).

Proof. Direct consequence of the graviton identification in HCSM-45 [7]. The graviton multiplet is $54B_1 \oplus 54B_2$ in $\text{Sym}^2(\Omega^1|_{\lambda=4})$, which is the image of the bilinear map $\omega \otimes \omega \rightarrow \text{Sym}^2(\Omega^1)$ restricted to the $B_1 \oplus B_2$ sector. $\square$

The nonlinear self-coupling of the graviton is generated by the composite structure of the bilinear $h = \omega \otimes \omega$, combined with the Gaussian path integral over ω . The effective action for h is obtained by

$$e^{-S_{\text{eff}}[h]} = \int \mathcal{D}\omega \delta(h - \omega \otimes \omega) e^{-S[\omega]}, \quad (24)$$

and its expansion in powers of h generates the graviton self-couplings. The D_4 structure of the effective vertices is determined by the D_4 structure of the composite map $\omega \otimes \omega \rightarrow h$, which is a D_4 -equivariant bilinear. The combination produces the derivative-included vertices counted in Theorem 4.2.

7.3 Faddeev–Popov field-independence

Proposition 7.2 (Field-independent FP measure). *The Faddeev–Popov measure on the discrete torus does not contribute to nonlinear vertices.*

Proof. In the Faddeev–Popov derivation of the Newton constant (HCSM-48 [9]), the gauge-fixing condition on the discrete torus is $d^*\omega = f$, linear in ω . The Faddeev–Popov operator is the scalar Laplacian on Ω^0 , whose determinant is field-independent. Therefore the Faddeev–Popov measure does not contribute to the nonlinear vertices on the discrete torus. The nonlinearity of the effective gravitational action arises entirely from the composite structure (24). $\square$

Remark 7.3 (Consequence for the composite-operator effective action). The composite-operator effective action — the ω -integral (24) and its expansion in powers of h — is the subject of HCSM-50 [11]. The present paper establishes the D_4 -invariant content of the vertex spaces on which that effective action acts.

8 Inputs and derived

8.1 Derived

1. The pure-polynomial A_1 counts $(0, 2, 0, 3)$ at orders 1, 2, 3, 4 (Theorem 3.2).
2. The derivative-included A_1 counts $(5, 12, 11)$ at orders 2, 3, 4 with two derivatives (Theorem 4.2).
3. The odd-derivative selection rule (Theorem 5.1).
4. The $O(3, 1)$ embedding (Theorem 6.2).
5. The composite bilinear identification (Proposition 7.1).
6. The Faddeev–Popov field-independence (Proposition 7.2).

8.2 Inputs

1. The graviton polarization space $V = B_1 \oplus B_2$ (HCSM-45 [7]).
2. The D_4 character table and tensor-product rules (HCSM-05 [5], HCSM-06 [6]).
3. The $\ker D \cong 2A_1 \oplus E$ decomposition (HCSM-03 [4]).
4. The cycle-index formula (3) and the orthogonality relation (8). These are standard representation-theoretic facts.
5. The Faddeev–Popov gauge-fixing condition $d^*\omega = f$ (HCSM-48 [9]).

8.3 What is not claimed

We do not claim to:

- Prove the exact masslessness of the graviton. This is the subject of HCSM-46 [8].
- Derive the Faddeev–Popov normalization or the Newton constant. These are the subjects of HCSM-48 [9] and HCSM-49 [10].
- Compute the full composite-operator effective action. This is the subject of HCSM-50 [11].

This paper establishes the D_4 -invariant content of the graviton vertex spaces — the pure-polynomial and derivative-included counts $(0, 2, 0, 3)$ and $(5, 12, 11)$, the odd-derivative selection rule, and the $O(3, 1)$ embedding.

9 Reproducibility

All numerical claims are verified at machine precision in IEEE-754 double precision, and are exact in rational arithmetic. The verification script is provided in the HCSMcompanion repository at `12-VERIFICATION/HCSM-47_verification.py`. It constructs the D_4 character table, computes $\text{Sym}^n(V)$ for $n = 1, 2, 3, 4$, computes the derivative-included vertex spaces $\mathcal{V}_{n,2}$ for $n = 2, 3, 4$, and verifies:

1. The pure-polynomial A_1 counts $(0, 2, 0, 3)$.
2. The derivative-included A_1 counts $(5, 12, 11)$.
3. The dimension checks $\dim \text{Sym}^n(V) = n + 1$ for $n = 1, 2, 3, 4$.
4. The dimension checks $\dim \mathcal{V}_{2,2} = 22$, $\dim \mathcal{V}_{3,2} = 38$, $\dim \mathcal{V}_{4,2} = 54$.
5. The odd-derivative vanishing for $m = 1, 3$.
6. The $O(3, 1)$ embedding via the $\ker D$ decomposition.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSMpaper stack.

10 Conclusion

This paper has established the self-coupling structure of the graviton in the Hodge Complex Standard Model.

The pure-polynomial D_4 -invariant content of the graviton vertex spaces is $(0, 2, 0, 3)$ at orders 1 through 4. Pure-polynomial kinetic and cubic graviton self-couplings are forbidden by D_4 symmetry — the discrete analogue of the standard statement that gauge invariance forbids pure-polynomial self-couplings of the graviton.

The derivative-included content with two derivatives is $(5, 12, 11)$ at orders 2, 3, 4. These establish the representation-theoretic content required for the kinetic, cubic, and quartic vertices. The pattern reproduces the derivative structure of the Einstein–Hilbert expansion.

The odd-derivative selection rule forbids self-couplings with an odd number of derivatives, as a consequence of the faithful action of D_4 on the derivative irrep E .

The D_4 action on $\ker D \cong 2A_1 \oplus E$ embeds as a finite subgroup of $O(3, 1)$, so D_4 -invariant structures are a subset of Lorentz-invariant structures. The vanishing of pure-polynomial kinetic and cubic content is therefore a statement about the full Lorentz-invariant structure, not only about the discrete subgroup.

The graviton is a composite bilinear $h = \omega \otimes \omega$ in the substrate 1-form ω , and its nonlinear self-coupling is generated by the composite structure through the effective action

$$e^{-S_{\text{eff}}[h]} = \int \mathcal{D}\omega \delta(h - \omega \otimes \omega) e^{-S[\omega]}. \quad (25)$$

The Faddeev–Popov measure does not contribute nonlinear vertices on the discrete torus because the gauge-fixing condition is linear and the determinant is field-independent.

The explicit computation of the composite-operator effective action, the reduction of the raw representation-theoretic counts to physical vertex counts, and the matching to the Einstein–Hilbert expansion are stated as the remaining open steps toward the full nonlinear Einstein equations (HCSM-50 [11]).

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A Web Crawler Metadata

A.1 Dublin Core

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DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.date.issued: 2026-09
DC.type: Text / Preprint / White Paper
DC.format: application/pdf; text/x-latex
DC.language: en
DC.publisher: Entropia Research Institute
DC.rights: All rights reserved
DC.subject: Hodge Complex Standard Model; HCSM; graviton self-coupling;
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composite operator; Faddeev-Popov field-independence
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forbidding pure-polynomial kinetic and cubic self-couplings; the
derivative-included counts with two derivatives are (5, 12, 11) at
orders 2, 3, 4, establishing the representation-theoretic content
for the kinetic, cubic, and quartic vertices. The odd-derivative
selection rule forbids odd-derivative self-couplings. The D4 action
on  $\ker D = 2A_1 + E$  embeds as a finite subgroup of  $O(3,1)$ .
DC.identifier: HCSM-47-2026-09

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A.2 Schema.org JSON-LD

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A.3 BibTeX for Automated Indexing

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  note = {ORCID: 0009-0004-5445-1744},
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    the self-coupling structure: the pure-polynomial  $D_4$ -invariant
    counts are (0, 2, 0, 3) at orders 1 through 4, forbidding
    pure-polynomial kinetic and cubic self-couplings; the derivative-

```

included counts with two derivatives are (5, 12, 11) at orders 2, 3, 4. The odd-derivative selection rule forbids odd-derivative self-couplings. The D4 action on $\ker D = 2A_1 + E$ embeds as a finite subgroup of $O(3,1)$.}

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